Hemant Banke

M.Stat., Indian Statistical Institute, Kolkata | Goldman Sachs | Quantitative Strategist
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Academic Profile

Statistics-trained quantitative researcher seeking doctoral study in reinforcement learning, with interest in model-based multi-agent RL. Earned an M.Stat. from the Indian Statistical Institute, Kolkata, achieving First Division with Distinction, followed by a vast Quant and ML experience at Goldman Sachs and Wells Fargo. Research and applied work spans RL, MARL, Stochastic processes, Monte Carlo and MCMC methods, optimization, causal inference, preference modeling, etc.

Research Interests: Reinforcement learning, Multi-Agent RL, Model-based RL, Differentiable planning, Spatial beliefs

Education

Indian Statistical Institute (ISI Kolkata) Sept 2021 – June 2023
Master of Statistics (M.Stat.), Specialization in Computational Statistics
Academic Standing: First division with **Distinction**; Received full tuition waiver with stipend
Selected Coursework: Stochastic Processes, Probability Theory, Pattern Recognition, Statistical Inference, Multivariate Analysis, Statistical Computing, Theory of Games and Statistical Decisions, Time Series Analysis

Holkar Science College (affil. Devi Ahilya Vishwavidyalaya) July 2018 – June 2021
Bachelor of Science (B.Sc.), Major in Statistics; Minors in Mathematics and Computer Science
Academic Standing: First division with **Distinction**; Won full scholarship

Research Experience

Multi-Agent RL for Defensive Coordination in Soccer Ongoing, 2026
with Soham Majumder, Leuphana University of Lüneburg

- Formulating defensive positioning as a two-team zero-sum Markov game, aiming to be the first framework to optimize defensive trajectories under co-adapting attackers. Designing a **graph-transformer policy** over spatiotemporal player graphs using **factorized space-time attention**, to handle variable player counts and play durations, while keeping architecture aligned with **CTDE**.
- Developing an **offline-to-online** training pipeline (IQL pre-training followed by adversarial self-play refinement with trust-region constraints) on a physics-based reward, with per-agent **counterfactual credit assignment** and theoretical grounding via the **simulation lemma** and concentrability.

RL for Goal-Based Wealth Management with Debt Decisions Ongoing, 2026
with Justin Morris Krieger, Wells Fargo and Olin Business School at Washington University

- Solving the GBWM stochastic optimal control problem via RL; extends the standard formulation to heterogeneous asset and debt accounts with voluntary debt decisions, thereby enlarging the state & action space beyond prior literature.

Concept Drift and Domain Transfer for GPP Estimation | Master's Thesis, ISI Kolkata 2023
Supervisor: Prof. Dr. B. Uma Shankar, Machine Intelligence Unit (Ex-Head), ISI Kolkata
Links: Thesis Report | Presentation

- Studied transfer of supervised models trained on global ecological data, to a low-data target domain (Indian terrain) **without ground truth**. Quantified virtual concept drift with **Drifter algorithm** using a distance measure between subset regressors, and estimated loss in target domain via Direct Loss Estimation using a **Child-Nanny** architecture.

Nowcasting Stock Implied Volatility with Twitter | Extension 2023
Supervisor: Prof. Dr. Diganta Mukherjee, Applied Statistics Unit, ISI Kolkata
Links: Project Report

- Extended Dierckx et al. (2022). Day-ahead IV forecasting gains via gradient-boosted ensembles combining Twitter derived attention and sentiment features with market-data features.

Professional Experience

Goldman Sachs May 2026 – Present
Associate, Quantitative Strategist | AWM Multi-Asset Solutions

- Engineering solutions for proprietary long-term wealth-management system supporting automated trading and analytics of **$510B** AUM across multiple asset classes.

Tools and Methods: Python; TypeScript; Slang; Optimization

Wells Fargo June 2023 – May 2026
Quantitative Analytics Specialist | Consumer Model Development Center

- Designed reward functions for a high-dimensional RL problem with an **axiomatic approach**: specified desired optimal properties, derived the class of functions they induce, selected a functional form within it, and analyzed its tradeoffs against planning variables.
- Built a **Bradley–Terry–Luce** preference model to rank next-best advice for customers conditioned on their goal plan.
- Solved multi-goal capital allocation as a LP problem with heterogeneous accounts and debt decisions; sampled within 1800+ dimensional feasibility region using **Hit-and-Run MCMC** sampling in a lower-dimensional latent space.
- Built a high-concurrency **Monte-Carlo** simulation framework (Python, OOP, multiprocessing) to backtest allocation strategies over correlated geometric Brownian motion asset paths.
- Designed a **synthetic-data pipeline** for high-complexity customer profiles under conditional correlations and strict edge cases, improving training density for downstream financial models.
- Estimated heterogeneous treatment effects of digital interactions through **Causal inference** (T-Learner with GBM, propensity-score matching).
- Taught and designed 18+ hours of theoretical ML curriculum to 80+ analysts, later used for analyst onboarding.
- **Leadership**: Found and led a team of 4 to develop first in-house Institutional RAG; Mentored 2 interns on Reward-design projects.

Recognition: Two Quarterly Recognitions and multiple Spotlight Awards
Tools and Methods: Python; PySpark; SciPy; Optimization; Stochastic Processes; Monte Carlo and MCMC Methods; Preference Models; Causal Inference; RL; Reward Design; GBWM

Selected Project

Ensemble Methods for Expected Goals (xG) Modeling
Supervisor: Dr. Deepayan Sarkar, Stat-Math Unit, ISI Delhi
Links: Project Report

- Studied Ensemble-learning methods and used **Stacked ensemble** (penalized logistic + kNN base models and GBM meta-learner) for shot-quality estimation from spatial and contextual features, achieving 78.5% ROC-AUC.
- Addressed class imbalance using ROSE. Motivated further study through gaussian smoothing of spatial variables.

Checkout more projects here .

Technical Skills

- **ML & RL**: Policy Gradients; Actor-Critic Methods; Self-Play; Offline and Offline-to-Online RL; Multi-Agent RL; CTDE; Graph Neural Networks (GATs, message passing); Deep Learning; Attention & Transformer Architectures; Preference Learning (Bradley–Terry); Concept-Drift Estimation; Loss Estimation
- **Statistical & Computational Methods**: Game Theory; Monte Carlo & MCMC; Optimization (LP, MILP, Convex); Statistical & Causal Inference; Stochastic Processes; MDPs; GBWM
- **Programming**: Python (PyTorch, PySpark, NumPy, pandas, SciPy, Matplotlib); R; C++; SQL; Git; Linux

Honors and Achievements

- **All India Rank 39** in ISI M.Stat. 2021 exam; **All India Rank 84** in IIT-JAM Mathematical Statistics 2021 exam
- Scored **Top 0.1%** Rank in IIT-JEE Main 2018 exam (1+ million candidates)
- Full tuition waiver with stipend (Master's) and full scholarship (Bachelor's); graduated both with Distinction
- YC Startup School 2026; Won **$25K** worth AI credits
- Ranked **globally 1st** and **19th** in Manual Trading rounds, IMC Prosperity 4 trials
- Won and led Inter-school BalVigyaan CS Competition twice; Built HLSL skin shader with Subsurface scattering & IBL
- Finalist in Code-N-Counter 2020 Hackathon, organized by Nagarro and Coding Ninjas

Extracurricular Activities

- **Teaching**: Prepared exhaustive ML curriculum and taught to analysts at Wells Fargo, covering theory behind all key traditional ML algorithms and concepts (2025); Doubt Clearing Coach at Cheenta School of Statistics (2021)
- **Social Responsibility**: CSR activities and donations supporting education of underprivileged children; contributed to cancer-awareness outreach through GCCI's nationwide programme
- **Research and Technical Communities**: Cohere Labs Open Science Community (2026); E-Cell (2021); Google Developer Group, Facebook Developer Community (2018)
- Built a Game Engine with C++ & OpenGL at age 11; started coding shaders and selling them on Unity3D Asset Store
- Started Freelancing web-apps during Bachelor's and was hired as a founding engineer for a startup; developed MVP of a two-sided service marketplace with propriety Escrow and Wallet systems supporting cards and crypto

Training and Certifications

ML Summer School, Cohere Labs (2026); YC Startup School (2026); Akuna Capital Options 101 (2025); Computer Programming for Engineers, Wichita State University (2019)